

EduGenie – Google Gemini Powered Learning Assistant

Code Layout, Readability and Reusability

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Recommended / Existing Logical Layout

Component	Purpose	Reusability Practice
FastAPI application entry point	Starts server and registers routes	Centralized application setup
Feature modules	Q&A, explanation, summary, quiz, learning path	Each feature has focused responsibilities
Gemini Service Layer	Model discovery, selection, fallback, generation	Shared service avoids duplicated API logic
Templates	Jinja2 web pages	Reusable presentation structure
Static assets	CSS/JavaScript	Separate UI concerns from backend logic
Configuration	Environment variables / dotenv	Secrets separated from source code
Tests	pytest test cases	Repeatable verification of behavior

Code Quality Principles

Use descriptive names, small functions, clear module boundaries, reusable service functions, environment-based configuration, input validation, and comments/docstrings where behavior is non-obvious.